DTIGCCN: Prediction of drug-target interactions based on GCN and CNN

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Abstract—Drug-target interaction (DTI) prediction plays an important role in drug repositioning, drug discovery, and drug design. In recent years, some DTI prediction methods based on machine learning have been proposed. They usually extract features from chemical genomics data. However, these methods are easy to extract redundant information that is not fully related with the prediction task and ignore the latent relationship between drug and target. This paper presents a new DTI prediction model named DTIGCCN. The model uses a spectral-based graph convolutional network (GCN) to extract features from drug and target expression profiles respectively, and a convolutional neural network (CNN) to extract latent associations between drug and target. Finally, the extracted features are concatenated together and fed into an effective classifier for prediction. The advantage of DTIGCCN is that the extracted features are more refined and targeted and the correlation between drug and target is fully applied to the prediction. Experimental results show that our model is superior to the conventional DTI prediction methods based on feature extraction and provides a new idea and method for DTI prediction.

Keywords—DTI prediction, feature extraction, GCN, CNN

I. Introduction

Predicting drug-target interactions (DTIs) is a key area for drug discovery and drug repositioning research [1-2]. Although there are various biotechnologies available for DTI prediction, considering the cost of biological experiments and research cycle, large-scale DTI prediction experiments still have many limitations.

The emergence of high-throughput screening (HTS) technology has brought new opportunities for the development of DTI prediction methods. Extensive experimental data on drugs and genes can be obtained from online databases, such as Drugbank [3]. Therefore, in recent years, many scholars have tried to study effective computational methods to identify new DTIs. These methods can be roughly classified into three categories. The first is structure-based methods [4]. These methods usually require a three-dimensional (3D) structure of the protein, and for those proteins whose structure is unknown, their performances are limited. The second type is based on the ligand similarity [5]. The basic assumption of this type of methods is that the ligand proteins with similar structure have similar biological activity, that is, they will react with the same type of target. The disadvantage of these methods is that they cannot reliably predict new drugs without knowing any target information. The third is machine learning and deep learning methods based on chemical genomics. Support vector machine (SVM), random forest (RF) and other classical machine learning models have been widely used to predict DTIs [6-7]. However, SVM is difficult to train on large-scale training samples, and RF is easy to cause overfitting when classifying noisy data. In addition, deep learning methods are also widely used in the construction of DTI prediction models. DTINet [8] uses the matrix-completed method to make DTI predictions. However, the unsupervised training method separates feature processing and prediction, which may not produce the best prediction results. A domain-adversarial multi-task framework was proposed to predict the potential purpose of small-molecule compounds [9]. Restricted Boltzmann Machines (RBM) [10], Deep-belief Network (DBN) [11], CNN-based DeepConv-DTI deep model [12], NegStacking [13] and deep neural networks [14] have been applied to DTI prediction. However, such methods rarely consider the latent relationship between drug and target, and as the network scale increases, redundant information will be extracted, which will affect the final prediction performance. In recent years, graph convolutional networks (GCNs) are the most popular deep learning methods for graphs and have been used to in silico drug discovery [15], prediction of drug-target binding affinity [16], exploring the relationship between drugs and diseases in biological networks [17], drug response prediction [18], and protein reaction interface prediction [19]. Although these methods have achieved good performances in the corresponding fields, they must have graph-structured data as input, and have difficulty processing high-dimensional feature vectors. In addition, these methods do not fully consider the latent association information between drug and target.

Aiming at the shortcomings of the existing feature-based DTI prediction methods, this paper defines the DTI prediction problem as a supervised binary classification problem and designs a new model named DTIGCCN. In this model, we use spectral-based graph convolutional network (GCN) to extract the features of drug and target expression profiles respectively and convolution neural network (CNN) to extract the latent association information of drug-target pair. Finally the above two types of features are concatenated and fed into the classifier for prediction. The main contributions of this paper are as follows:

- Extract features of drug and target respectively to avoid too many redundant features being used for subsequent classification and prediction;
- Coarsen the graph, remove unimportant features, and then extract features based on the graph to reduce invalid information;
- CNN is used to extract the latent association information between drug and target.

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We performed a ten-fold nested cross-validation experiment on the data set of seven cell lines for model analysis, and compared DTIGCCN with other comparable methods. Experimental results show that our model could achieve better results.

II. Data and Model

A. Data Source

The National Institute of Health (NIH) proposed the Library of Integrated Network-based Cellular Signatures (LINCS) project to develop biological labels on the drug characteristics at the level of cells. L1000 is a database obtained by using L1000 biotechnology in this project, which uses more than 20,000 small-molecule compounds to stimulate cell lines to obtain millions of genome-wide expression profiles, and uses the gene knockout method to knock out 4,000 genes perturbation data in their respective cell lines [20]. This provides a unified and extensive gene expression profile for DTI prediction. DrugBank is a comprehensive drug data source, which records the chemical, pharmacological and drug characteristics of more than 8,000 drugs [3].

We obtained the transcriptional response data of drug and target protein perturbations in seven cell lines (A375, A549, HA1E, HCC515, HEPG2, PC3, VCAP) from L1000 database and the relationship between drug and target from the DrugBank database. We treated each interacting drug-target pair as a positive sample and the combination of drug and non-target protein as a negative sample. The data set of each cell line used in this study is shown in Table I. In order to avoid the fact that too many negative samples lead the final training model to be more inclined to predict the sample as negative, for each cell line, we randomly and uniformly sampled negative samples to keep the ratio of positive samples to negative samples as 1:2. The drug and target expression profiles are 978 dimensional feature vectors, which are expressed as follows:

Definition 1: $\mathcal{T}_i^{(\alpha)}$ denotes the $i^{th}(1 \le i \le q)$ target expression profile of cell line α, which is represented as:

$$\mathcal{T}_i^{(\alpha)}=[t_{i,1}^{(\alpha)},t_{i,2}^{(\alpha)},\ldots,t_{i,k}^{(\alpha)},\ldots,t_{i,978}^{(\alpha)}] \qquad (1)$$

TABLE I. Data Volume of Each Data Set in Seven Cell Lines

	Drug	Target	Non-target	Positive samples	Negative samples
A375	520	363	2754	796	1432080
A549	525	366	2648	805	1390200
HA1E	533	372	2707	818	1442831
HCC515	471	334	2516	689	1185036
HEPG2	370	356	2520	557	932400
PC3	643	378	2866	963	1842838
VCAP	521	377	3003	809	1564563

where $t_{i,k}^{(\alpha)}$ is the $k^{th}(1 \le k \le 978)$ feature of $\mathcal{T}_i^{(\alpha)}$.

Definition 2: $\mathcal{D}_j^{(\alpha)}$ denotes the $j^{th}(1 \le j \le m)$ drug expression profile of cell line α, which is represented as:

$$\mathcal{D}_j^{(\alpha)}=[d_{j,1}^{(\alpha)},d_{j,2}^{(\alpha)},\ldots,d_{j,k}^{(\alpha)},\ldots,d_{j,978}^{(\alpha)}] \qquad (2)$$

where $d_{j,k}^{(\alpha)}$ is the $k^{th}(1 \le k \le 978)$ feature of $\mathcal{D}_j^{(\alpha)}$.

B. Model Framework

In order to fully extract the effective feature information, we used GCN to extract features from drug and target expression profiles, respectively. First, graphs are constructed based on the drug and target gene expression profiles, and each drug or target has a graph corresponding to it. Then the graph is coarsened and a balanced binary tree is created. Finally, convolution and pooling operations defined on the graph are performed. Two feature vectors f1 and f2 are obtained, which represent feature information extracted from the drug and target expression profiles respectively. In order to obtain the latent association between drug and target, the expression profiles of the drug and the target are vertically spliced into a 2×978 matrix, and CNN is used to extract features from the matrix to obtain the feature vector f3. Finally, f1, f2, and f3 are concatenated and fed into the classifier for prediction. If the result of the classification is a positive sample, the drug-target pair is predicted to have interaction; otherwise, no interaction is predicted between them. The structure of the entire model is shown in Fig. 1. Next, we will introduce the key methods used in the model.

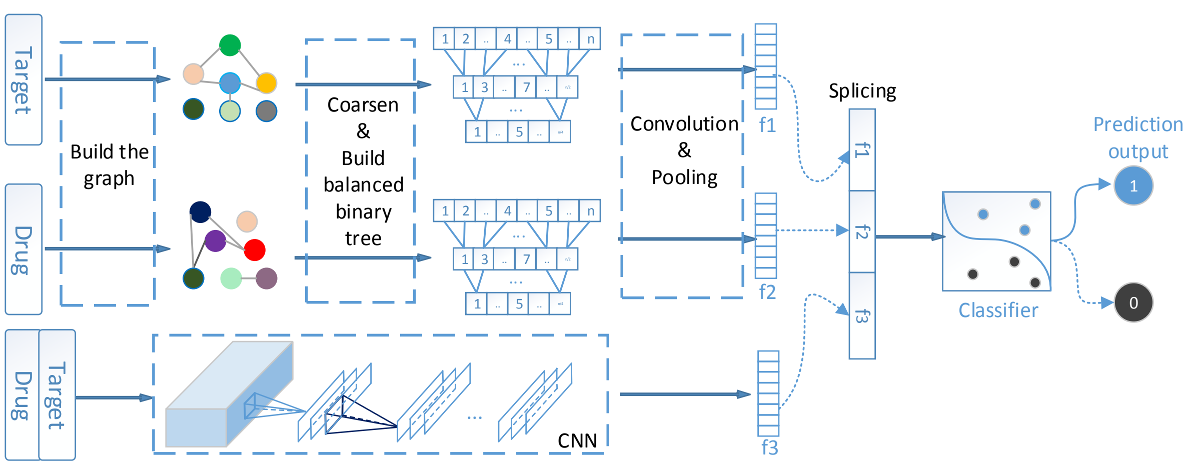

Fig. 1. DTIGCCN model structure

III. Methods

A. Graph Construction

In the DTIGCCN model, we first need to construct a graph for each drug and target respectively. Each drug and target is represented by a vector with 978 features. We regard each feature as a vertex in a graph, so each graph contains 978 vertices. Gaussian kernel function is used to calculate the connection strength $W_{i,j}$ between any two vertices i and j as shown in (3)-(4):

$$W_{i,j}=W_{j,i}=\begin{cases} e^{-\frac{|dist(i,j)|^2}{2\theta^2}} & if\ dist(i,j)\leq k \\ 0 & otherwise \end{cases} \quad (3)$$

$$\text{dist(i,j)}=\begin{cases} t_{l,i}^{(\alpha)}-t_{l,j}^{(\alpha)},\ \text{for}\ t_{l,i}^{(\alpha)},t_{l,j}^{(\alpha)}\in\mathcal{T}_l^{(\alpha)} \\ d_{l,i}^{(\alpha)}-d_{l,j}^{(\alpha)},\ \text{for}\ d_{l,i}^{(\alpha)},d_{l,j}^{(\alpha)}\in\mathcal{D}_l^{(\alpha)} \end{cases} \quad (4)$$

where $\theta > 0$ is the bandwidth of the Gaussian kernel and k is the threshold of strength.

Finally, an undirected graph $G(V, W, E)$ can be obtained for each drug and target expression profile, where V is a finite set of vertices and $|V| = 978$. W is the connection strength matrix. For two vertices i and j, if $W_{i,j} \neq 0$, $E\ (i,j)\ (1 \leq i,j \leq 978, i \neq j)$ is an edge of G. Fig. 2 gives a schematic example of how to construct a graph.

B. Graph Coarsening and Pooling

The pooling operation requires meaningful neighborhoods on the graph, so before doing the pooling operation, we need to cluster the vertices of the graph to bring similar vertices together. In this study, we use spectral clustering [21] to cluster the vertices of the graph. Since the pooling operation needs to be performed multiple times in multiple layers, we adopt the coarsening method of Graclus multi-level clustering algorithm [22] and use standardized cutting as the objective function [23] to obtain multi-level clustering results. Graclus uses a greedy algorithm to calculate a continuous multi-layer coarsened graph of a given graph, given an initial graph which is not marked on a certain layer, and then traverses each vertex in a random access manner. For node i, if it is unmarked and there are unmarked vertices in its neighborhood, select a vertex j that maximizes the local normalized cutting function $W_{i,j}(1/d_i + 1/d_j)$ from the neighborhood of i. Vertices i and j are combined and marked. d_i and d_j are the degrees of vertices i and j respectively. If all vertices in the neighborhood of i have been marked, then mark i. The above process is repeated until all the vertices are marked, and the coarsening of the current layer graph is finished. Algorithm 1 describes the coarsening process. As shown in Fig. 3(a), the original graph G_0 is coarsened by two layers to obtain G_1 and G_2.

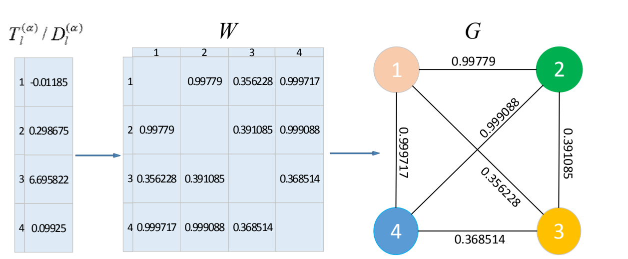

Fig. 2. The example of constructing a graph based on the drug/target expression profile

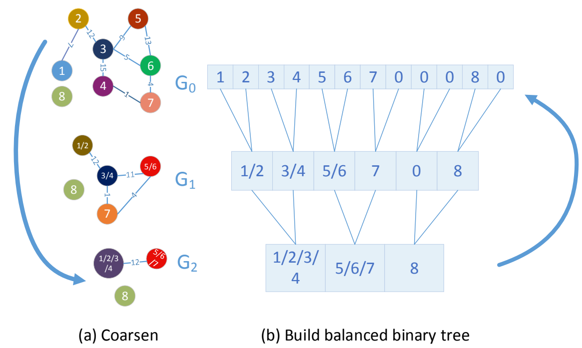

Fig. 3. Graph coarsening and the process of creating a balanced binary tree

Algorithm 1 coarsen

Input: $G_0 = (V, W, E)$, m

Output: coarsened graph set $G_{coarsen}$

for k=1 to m **do**
  for each $V_i \in V$ *and* V_i *is unmarked* **do**
    select vertex j according to $\max(W_{i,j}(1/d_i + 1/d_j))$;
    mark V_i, V_j;
  end for
  obtain G_k;
end for
return $G_{coarsen} = \{G_0, G_1, \ldots, G_m\}$;

After graph coarsening, the vertices in the coarsening results of each layer still exist in the form of a graph structure. In order to make the graph's pooling operation as effective as one-dimensional pooling, we need to rearrange the vertices [24]. The rearrangement process has two steps: (1) create a balanced binary tree; (2) rearrange the vertices. After coarsening, there are two forms of nodes in each layer graph: vertex with two child nodes (level 1 node) and vertex with one child node (level 2 node). Based on the coarsened result, starting from the most coarsened graph, the balanced binary tree is constructed layer by layer. During the construction process, some fake nodes need to be added. These fake nodes initialized with neutral values will not affect the final pooling operation. The following rules should

Algorithm 2 create balanced binary tree

Input: $G_{coarsen} = \{G_0, G_1, \ldots, G_m\}$

Output: balanced binary tree BT

for i=0 to m **do**
  for each $v \in V_i$ **do**
    if v has one child **then**
      create a fake node v_{fake} as the child of v;
      add v_{fake} to V_{i+1};
    else if v has no child **then**
      create fake nodes v_{fake1}, v_{fake2} as children of v;
      add v_{fake1}, v_{fake2} to V_{i+1};
    end if
  end for
end for

be followed when creating a balanced binary tree: a level 1 node has either two level 1 child nodes, or a level 1 child node and a level 2 child node; a fake node can only have two fake nodes as child nodes. In the end, the rearranged vertex sequence is at the bottom of the balanced binary tree, and the pooling operation on the rearranged one-dimensional sequence is more efficient. Algorithm 2 shows how to create a balanced binary tree, as illustrated in Fig. 3(b).

C. Graph Convolution Operation

For structured graph data, the traditional convolution kernel with finite size designed on regular grid data is no longer applicable. Therefore, designing a convolution kernel suitable for graph structure is one of the key steps for graph convolution [24]. First, based on the Laplacian matrix, a Fourier basis is constructed. The robust mathematical representation of the graph is a normalized Laplacian matrix, which is defined as:

$$L = I_n - D^{-\frac{1}{2}} A D^{-\frac{1}{2}} \tag{5}$$

where I_n is the identity matrix, A is the adjacency matrix, and D is the diagonal matrix, $D_{ii} = \sum_j A_{ij}$. Since L is a symmetric semi-definite matrix, it can be expressed as:

$$L = U \Lambda U^T \tag{6}$$

where $U = [u_0, u_1, u_2 \dots u_{n-1}] \in R^{N\times N}$ is the set of orthogonal eigenvectors of L, which is the Fourier basis, and $\Lambda = diag(\lambda_0, \lambda_1, \lambda_2 \dots \lambda_{n-1}) \in R^{N\times N}$ is the eigenvalue matrix of L. Then the convolution operation on the Fourier domain is defined as follows:

$$y = g_\theta(L)x = g_\theta(U\Lambda U^T)x = U g_\theta(\Lambda) U^T x \tag{7}$$

where $g \in R^N$ is the convolution operator, $x \in R^N$ is the expression profile of the drug and target, and y is the new feature extracted by graph convolution. Spectral-based graph convolution operators all follow this definition. Choosing different convolution filters will generate different operators. In this study, the convolution operator is defined as the Chebyshev polynomial of the eigenvalue diagonal matrix [25],

$$g_\theta(\Lambda) = \sum_{k=0}^{K-1} \theta_k T_k(\widetilde{\Lambda}) , \widetilde{\Lambda} = 2\Lambda/\lambda_{max} - I_n \tag{8}$$

Chebyshev polynomials are recursively defined as:

$$\begin{gathered} T_k(x) = 2xT_{k-1}(x) - T_{k-2}(x), \\ T_0(x) = 1, T_1(x) = x \end{gathered} \tag{9}$$

Finally, the convolution operation can be written as:

$$\begin{gathered} \mathrm{y} = g_\theta(L)x = \sum_{k=0}^{K-1} \theta_k T_k(\widetilde{L})x , \\ \widetilde{L} = 2L/\lambda_{\max} - I_n \end{gathered} \tag{10}$$

IV. Experiment and Analysis

In this section, the model's performance is evaluated by three assessment metrics, which are accuracy (ACC), F1-score and area under ROC curve (AUC). First, we conduct the ablation study on the model to illustrate the importance of each module. Then, the impact of different hyper-parameters on the model performance are analyzed to obtain the optimal model parameters by ten-fold nested cross-validation. Finally, the optimal DTIGCCN model is compared with other related methods. The parameters and settings used in the DTIGCCN model are listed in Table II.

TABLE II. The Parameters and Settings Used in GCN,CNN,DNN and DTIGCCN

Parameters	Settings
Number of convolution kernels	50
Convolution kernel size	30
Number of graph convolution filters	{[4,8],[8,16],[16,32],[40,80],[100,200], [200,400],[400,800]}
Size of graph pooling	[2,2]
Fully connected neurons	[200,10]
Epoch	100
Batch size	16
Dropout	0.5
Regularization	0.0005
Learning rate	0.001
Optimizer	Adam

A. Ablation Experiment

In this section, we analyze the components of the DTIGCCN model and demonstrate their impacts. We compare three models: GCN, CNN and DTIGCCN. The parameter settings of the three models are shown in Table II, where the number of graph convolution filters is set to [200, 400] .

It can be seen from Fig. 4 that DTIGCCN can obtain the best ACC and AUC values in all seven cell lines. Compared with the results of CNN, DTIGCCN has a clear superiority. This is because the GCN module in the DTIGCCN model extracts the effective feature information carried by the drug and the target, thereby improving the overall performance of the model. Compared with the GCN, which extracts the feature information of drugs and targets respectively, the superiority of DTIGCCN is more obvious. This shows that the latent association information between drug and target has an important impact on the prediction performance. In addition, CNN extracts the feature information of the drug-target pair by sharing the convolution kernel. This design of CNN reduces the number of

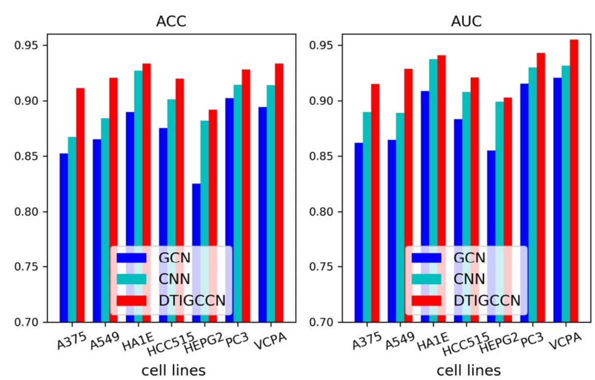

Fig.4. Effect of different model structures on experimental results

parameters, thereby reducing the risk of overfitting. The experimental results show that GCN and CNN have different effects on feature extraction. Combining the two modules together helps to extract more effective feature information, so that DTIGCCN can get better performance.

B. Parameter Analysis

The hyper-parameters play important roles in DTIGCCN, as they determine how the feature will be extracted. We conduct experiments to analyze the impacts of two key parameters through ten-fold nested cross-validation, i.e., the kernel function and the number of graph convolution filters. Fig. 5 and Fig. 6 respectively show the influence of the kernel function and the number of graph convolution filters on the experimental results. The horizontal axis represents different cell lines, and the vertical axis represents relatively increasing ratio (%). According to these two figures, we can find that:

Among the four kernel functions such as Laplace kernel (Lap), Exponent kernel (Exp), Gaussian kernel (Gau) and kernel defined in [24] (named as Gau 2), when the Gaussian kernel function is used, the model can obtain good experimental results in most cell lines. This is because the Gaussian kernel function is an excellent representative of the radial basis kernel function. It has good anti-interference ability for noise in the data, and has strong applicability and robustness to data with a wide range of values.

When the number of graph convolution filters on each layer is set to [200,400], the model can obtain better classification performance. This is because when the number of graph convolution kernels is too small, it may cause under-fitting, and too large may cause over-fitting, and eventually degrade the classification performance.

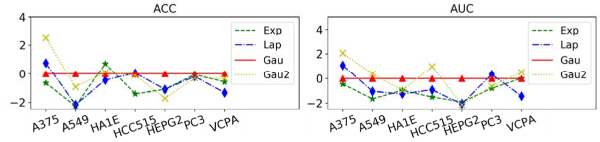

Fig. 5. The effect of different kernel functions

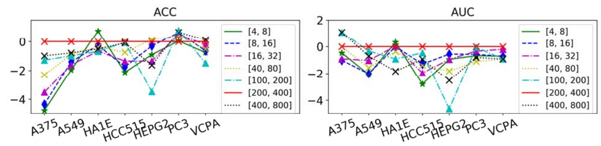

Fig. 6. The effect of the number of graph convolution filters

C. Comparison with Other Models

Based on the above experimental results, we found a set of optimal model configurations for DTIGCCN. Some parameter settings are shown in Table II. The Gaussian kernel function is selected to construct the graph, the number of graph convolution filters is set to [200,400], and DNN acts as the classifier. In this section, we will use the model built by this configuration to compare with other comparable models.

We compared DTIGCCN with several other models (Table III). Decision tree (DT) is a commonly used machine learning method with a tree structure; Random forest (RF) is an ensemble classifier using bagging technology, which integrates multiple

TABLE III. COMPARISON OF CLASSIFICATION PERFORMANCE BY USING DIFFERENT MODELS

Cell line	Model	DT	RF	DNN	DTI-RCNN	DeepDTA	chebNet	DTIGCCN
A375	ACC	0.7657	0.8661	0.8051	0.8024	0.8873	0.8523	**0.9113±0.0059**
	F1-score	0.6782	0.7808	0.7545	0.7857	0.8770	0.8916	**0.9321±0.0091**
	AUC	0.7470	0.8210	0.8061	0.8299	0.8959	0.8620	**0.9151±0.0139**
A549	ACC	0.7934	0.8595	0.8767	0.8409	0.8888	0.8650	**0.9207±0.0051**
	F1-score	0.6711	0.7344	0.8000	0.8219	0.8584	0.8908	**0.9458±0.0119**
	AUC	0.7603	0.7942	0.8443	0.8646	0.8949	0.8646	**0.9287±0.0089**
HA1E	ACC	0.7764	0.8862	0.8354	0.8444	0.9119	0.8898	**0.9334±0.0113**
	F1-score	0.6857	0.8056	0.7879	0.8333	0.9059	0.9177	**0.9548±0.0081**
	AUC	0.7598	0.8381	0.8358	0.8714	0.9373	0.9008	**0.9408±0.0086**
HCC515	ACC	0.8696	0.8986	0.7874	0.8208	0.9105	0.8754	**0.9198±0.0096**
	F1-score	0.7939	0.8142	0.6508	0.7852	0.8997	0.9067	**0.9457±0.0109**
	AUC	0.8538	0.8489	0.7469	0.8511	0.9167	0.8834	**0.9208±0.008**
HEPG2	ACC	0.8036	0.8750	0.7936	0.8250	0.8807	0.8252	**0.8919±0.0091**
	F1-score	0.7179	0.7586	0.6916	0.8191	0.8419	0.8668	**0.9273±0.0126**
	AUC	0.8006	0.8070	0.7751	0.8680	**0.9137**	0.8550	0.9028±0.0076
PC3	ACC	0.8685	0.8616	0.8596	0.8438	0.9138	0.9023	**0.9282±0.0073**
	F1-score	0.8119	0.7727	0.8137	0.8351	0.9154	0.9301	**0.9546±0.0071**
	AUC	0.8538	0.8173	0.8561	0.8677	0.9132	0.9153	**0.9430±0.0062**
VCAP	ACC	0.8189	0.9053	0.8195	0.7989	0.9242	0.8944	**0.9334±0.0103**
	F1-score	0.7067	0.8217	0.7397	0.7347	0.8736	0.9379	**0.9590±0.0088**
	AUC	0.7924	0.8542	0.8150	0.8140	0.9280	0.9207	**0.9550±0.0118**

decision trees; DNN[14] uses a deep neural network to predict DTI based on transcriptome data; DTI-RCNN [26] is a hybrid model used to predict DTI, which integrates LSTM network and CNN; DeepDTA [27] is a drug–target binding affinity prediction model with three-layer one-dimensional convolution; chebNet [24] uses a spectral-based graph convolution method for feature extraction. For a fair comparison, we follow most of the default parameter settings of the comparison model and adjust some parameters to obtain optimal results.

As shown in Table III, we can find that DTIGCCN is significantly better than several other methods in terms of ACC, AUC and F1-score. For traditional machine learning methods such as DT and RF, the original data may contain a certain amount of noise, which will cause over-fit problem and affect the classification performance. In addition, DT and RF do not consider the latent association information between drug and target. For DNN and DTI-RCNN models that predict DTI, DTIGCCN also has obvious advantages. Because the original data does not have a sequence relationship, DTI-RCNN cannot extract effective features using LSTM; DNN directly concatenates drug and target expression profiles and uses deep neural network to predict DTI. DeepDTA uses three-layer one-dimensional convolution for feature extraction of drugs and targets respectively; chebNet only uses GCN to extract feature information of drugs and targets. DNN, DeepDTA and chebNet do not consider the latent association information between drug and target, so the performances are also poor. We believe that the DTIGCCN model avoids extracting redundant feature information and captures the latent association information between drug and target, and makes the features used for classification more comprehensive and effective, thereby achieving better and more stable classification performance.

V. Conclusion

In this study, we define the DTI prediction problem as a binary classification problem, and design a hybrid model that combines spectral-based GCN and CNN to extract features of drug and target expression profiles. Based on this, a classifier is used to classify the extracted features. Compared with other methods, DTIGCCN can not only effectively extract the feature information of drug and target respectively, but also fully take into account the latent association between drug and target. The experimental results show that choosing the appropriate model structure will have an important impact on the final experimental results. For our model, the Gaussian kernel function is very helpful to construct a high-quality graph, which provides a solid foundation for the follow-up operations of feature extraction, so as to obtain better classification performance. Choosing appropriate graph convolution settings can improve the effectiveness of feature extraction.

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